Employee Attrition Risk Analysis & Workforce Optimization

Business Insights & HR Recommendations

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Tools: Excel | SQL | Python | Power BI

1. Business Problem Statement:

Employee attrition results in increased recruitment costs, productivity loss, and disruption to organizational continuity. Organizations often lack early warning mechanisms to identify employees at risk of leaving. This project focuses on analysing employee behaviour and workforce data to identify key drivers of attrition, segment employees by risk level, and support data-driven HR decision-making for improved retention and workforce optimization.

2. Data & Tools Used:

1. Employee HR dataset containing satisfaction, workload, promotion, tenure, and attrition status
2. Data Cleaning & Analysis: Excel, SQL, Python (Pandas, NumPy)
3. Data Visualization & Storytelling: Power BI
4. Risk Segmentation using business rule–based scoring

3. Key Business Insights:

- The organization shows a notable attrition rate, indicating a potential retention risk.
- Employees with **low satisfaction levels** are significantly more likely to leave.
- High workload (average monthly hours > 200) is strongly associated with attrition.
- Employees without promotion in the last 5 years show higher disengagement and exit rates.
- Certain departments have a higher concentration of **high-risk employees**, requiring targeted intervention.
- Risk segmentation (Low, Medium, High) effectively highlights employees with higher attrition probability.

4. Power BI Dashboard Insights:

Page 1 – HR Overview

- Provided a high-level view of total employees, attrition rate, and department-wise distribution.
- Enabled quick understanding of overall workforce health.

Page 2 – Attrition Drivers

- Identified satisfaction, workload, promotion status, and salary as key attrition drivers.
- Highlighted behavioural patterns contributing to employee exits.

Page 3 – Risk Segments

- Classified employees into Low, Medium, and High-Risk segments.
- Enabled HR teams to identify employees requiring immediate attention.

5. HR Recommendations:

- Implement regular employee engagement and satisfaction surveys.
- Monitor and manage excessive workloads to reduce burnout.
- Introduce structured promotion and internal mobility programs after 3–4 years of tenure.
- Focus retention strategies on high-risk departments.
- Use risk segmentation to proactively engage employees before attrition occurs.

6. Business Impact:

- Early identification of high-risk employees can significantly reduce attrition costs.
- Proactive HR interventions improve employee retention and workforce stability.
- Data-driven workforce planning supports long-term organizational growth.

7. Conclusion:

This project applies structured data preparation, exploratory analysis, and interactive visualization to uncover the behavioural and operational drivers of employee attrition. Using a combination of spreadsheet analysis, database querying, statistical analysis, and dashboard-based storytelling, the solution identifies high-risk employee segments and translates analytical findings into actionable HR insights. The outcome enables HR teams to shift from reactive attrition management toward proactive, data-driven workforce optimization and retention planning.